

Arnav Tiwari

Software Engineer | AI Systems & Full-Stack Development

Node.js • Django • PostgreSQL • REST APIs • LLM Application Development • React/Angular

arnavt292@gmail.com | [8329846328](tel:8329846328) | linkedin.com/in/arnav-tiwari-063278253 | github.com/Arnav10090 | [Portfolio](#)

PROFESSIONAL SUMMARY

Full Stack Software Engineer building production AI systems, REST APIs, and data-driven platforms across industrial and security domains. Experienced designing backend services in Node.js, Express.js, and Django, integrating LLM-powered agents and retrieval pipelines into production workflows, and owning features end-to-end from data model to deployment. Comfortable operating in ambiguous, safety- and security-relevant environments where correctness and reliability matter as much as speed.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript

Backend & APIs: Node.js, Express.js, Django, FastAPI, REST, GraphQL, JWT, OAuth 2.0, WebSockets

AI Engineering: LLM Application Development, Google Gemini API, Groq SDK, LangChain, Multi-Agent Systems, Embedding Models, Prompt Engineering

Databases: PostgreSQL, MongoDB, MySQL, SQLAlchemy, Drizzle ORM

Cloud / DevOps: Docker, Kubernetes, CI/CD, GitHub Actions, Git, Linux, Bash, Vercel

Frontend: React.js, Next.js, Angular, Redux, Tailwind CSS

CS Fundamentals: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

PROFESSIONAL EXPERIENCE

Software Developer Intern

July 2025 – July 2026

Hitachi India Pvt Ltd. | Nagpur, Maharashtra

- Designed and built an AI-powered Technical Specification generation platform with **30+ configurable document sections**, retrieving relevant context from prior specification data and using section-aware prompting to generate standardized engineering documentation across multiple industrial solution types.
- Engineered SLD visualization modules for steel plant electrical layouts, enabling real-time monitoring of motors and critical equipment, as part of a broader initiative that contributed to a **33% reduction in motor failures** and **54% reduction in hazardous incidents** across the plant.
- Built and shipped **5** internal React/Django web applications for electrical and steel equipment analysis (design review, testing/commissioning, and inspection workflows), replacing manual spreadsheet-based processes and improving team workflow efficiency by **43%**.
- Designed and automated a CRS document generation pipeline using n8n to orchestrate data collection and templated report assembly, reducing turnaround from **2 weeks to 2 days (80% faster)** and enabling delivery of **7–10 project documents weekly** that previously took a month.

Full Stack Developer Intern

May 2025 – July 2025

Payatu Security Consulting Pvt. Ltd. | Pune, Maharashtra

- Developed 15 REST API endpoints and debugged and optimized 8 using indexing, query optimization, and caching on a Node.js + Express.js backend, improving response time by **40%** in collaboration with cross-functional teams.
- Built a real-time logger handling **1,000+ log events/minute** to support audit tracking and access accountability across a 4-role RBAC system (Admin, System Engineer, Technician, Employee) for a WiFi admin dashboard managing **1,000+ users**.
- Built responsive Next.js + SCSS interfaces reducing page load time from **3s to 1.5s (50% faster)**, improving UI maintainability and user experience.

PROJECTS

SkillBridge – AI Adaptive Onboarding Engine

[Link](#) | [GitHub](#)

Python, FastAPI, React, SQLite, Docker, Ollama

- Architected an AI-adaptive onboarding engine using FastAPI and React, orchestrating a **6-stage asynchronous processing pipeline** with real-time status polling to evaluate candidate readiness and generate personalized learning roadmaps.
- Engineered a **3-layer skill matching pipeline** combining hybrid LLM-based skill extraction, skill-family normalization, and custom **384-dimensional Blake2b hashing embeddings** for deterministic semantic similarity search.
- Designed an original **Weighted Graph Traversal (WGT)** algorithm implementing Kahn’s Topological Sort, difficulty ramping, and P-score prioritization to generate prerequisite-aware learning pathways across a **71-skill taxonomy**.
- Integrated demand weights derived from **2,277 job descriptions** with a taxonomy enriched using **2,484 resumes**, restricting recommendations to a curated **58-module** catalog for deterministic roadmap generation.

Autopilot AI – AI Powered Project Planning Platform

[Link](#) | [GitHub](#)

Node.js, Express.js, PostgreSQL, React, Next.js, Groq SDK, OAuth 2.0, Ajv

- Architected a multi-agent AI backend using Node.js, Express.js, and PostgreSQL, orchestrating **4 specialized AI agents** through an **18-endpoint REST API** to automate requirement analysis, technology recommendation, risk assessment, and project planning.
- Engineered a fault-tolerant LLM orchestration layer utilizing strict JSON schema validation (Ajv), deterministic validation, and structured fallback mechanisms to improve reliability of multi-agent execution.
- Developed a task planning engine generating **35–50 engineering tasks** distributed across **6 technical modules**, producing dependency-aware execution plans, timeline estimates, and automated risk assessments.
- Implemented secure authentication using OAuth 2.0 and bcrypt-backed local authentication with session management, alongside runtime observability for monitoring multi-agent execution.

ContentForge AI – AI Content Generation SaaS

[Link](#) | [GitHub](#)

Next.js, TypeScript, PostgreSQL, Drizzle ORM, Neon, Groq API, Razorpay, Clerk

- Architected a production AI SaaS platform using Next.js, React, PostgreSQL (Neon), and Drizzle ORM to support scalable AI content generation, persistent storage, and multi-user session management.
- Engineered a low-latency LLM inference pipeline using Next.js Server Actions and the Groq API to power **18 specialized AI content generation templates** with structured prompt orchestration.
- Designed a usage monitoring and monetization backend implementing **7 REST API endpoints** for Razorpay subscriptions, AI generation history tracking, and authenticated user management via Clerk.

ACHIEVEMENTS

- Rated **5-star (rating 2029)** on CodeChef, ranking among top competitive programmers globally | Solved **400+ DSA** problems on LeetCode.
- 1st Runner Up – ArtPark Codeforge Hackathon** (Unstop) – competed against teams across India in a software development challenge.
- Finalist (Top 2.6% of 31,000+ Teams)**, OpenEnv Hackathon × Scaler School of Technology – Advanced through multiple rounds to the 36-hour onsite finals in Bangalore (Apr 2026).

EDUCATION

Indian Institute of Information Technology Nagpur (IIIT Nagpur)

November 2022 – July 2026

BTech in Computer Science Engineering | **CGPA: 7.58/10**